

Retail Sales Performance Analytics Dashboard

Built with Microsoft Power BI

Field	Details
Name	Sai Abhisek Sahoo
Roll Number	2305325
Batch/Program	Power BI Analytics
Submission Date	April 21, 2026
Tool	Microsoft Power BI Desktop

1. Project Title

Retail Sales Performance Analytics Dashboard using Microsoft Power BI

This capstone project presents an end-to-end business intelligence solution for a retail organisation operating across multiple Indian states. Using Microsoft Power BI, raw transactional data is transformed into an interactive five-page dashboard enabling data-driven decision-making at strategic and operational levels.

2. Problem Statement

Modern retail businesses generate hundreds of transactions daily across geographies and product lines. Without proper data visualisation, key business questions remain unanswered, leading to poor inventory decisions, missed targets, and declining customer satisfaction.

- No visibility into regional performance versus quarterly targets.
- No product-level profitability insight across categories and sub-categories.
- Delayed trend detection — seasonal dips and surges are identified weeks too late.
- Customer segmentation gap preventing differentiated marketing strategies.
- Scattered data sources with no unified cross-functional view.

3. Solution and Key Features

A five-page interactive Power BI dashboard addresses each challenge. The solution uses a star-schema data model with two tables, a Calendar dimension, and 18 DAX measures covering aggregations, profit margins, time-intelligence (YTD, MoM, YoY), and dynamic KPI labels.

Page	Name	Key Visuals
1	Executive Summary	KPI Cards, Monthly Trend, Sales vs Target Gauge
2	Regional Analysis	India Bubble Map, Region Bar, Donut Chart
3	Product Performance	Category Bar, Heatmap, Top-10 Sub-Categories
4	Customer & Segment	Segment Pie, Profit Bar, Top 10 Customers
5	Time Intelligence	YTD Line, MoM Waterfall, YoY Comparison

4. Tech Stack

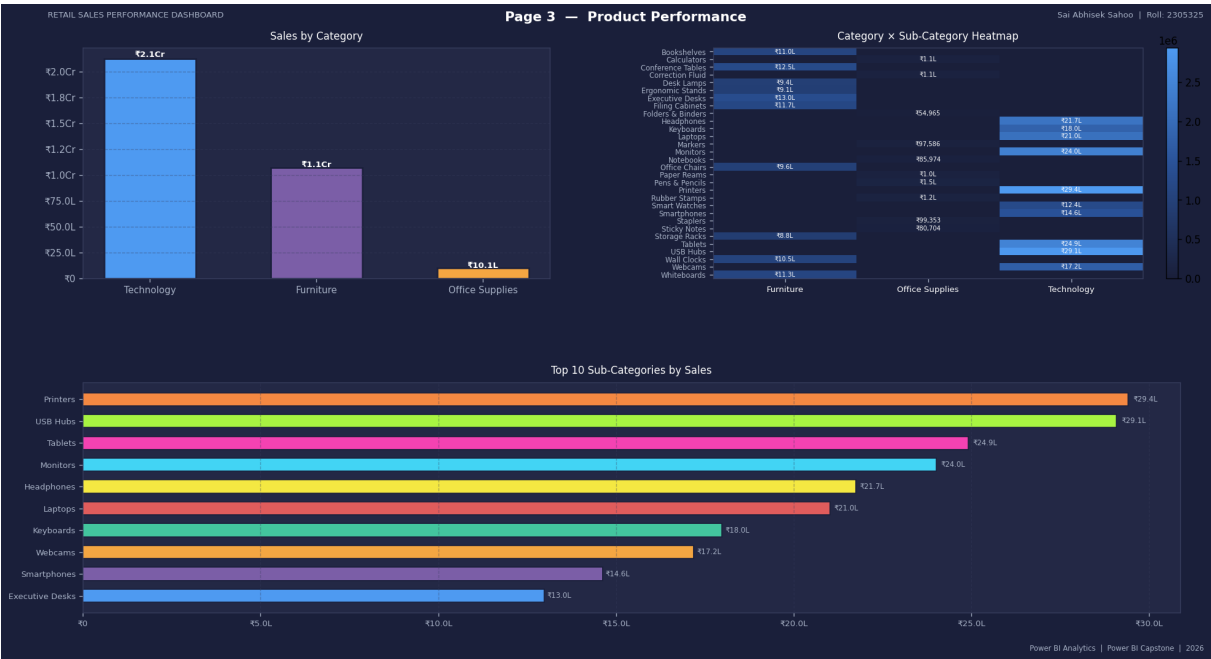
Component	Details
BI Tool	Microsoft Power BI Desktop
Data Format	CSV — 2 tables (500 sales rows + 96 target rows)
Data Modelling	Star Schema — Fact + Dimension tables
Query Language	Power Query (M Language) for data transformation
Formula Language	DAX — 18 measures including time-intelligence
Geography	Pan-India — North, South, East, West regions
Time Coverage	January 2023 – December 2024 (2 full years)
Version Control	Git / GitHub

5. Dashboard Screenshots

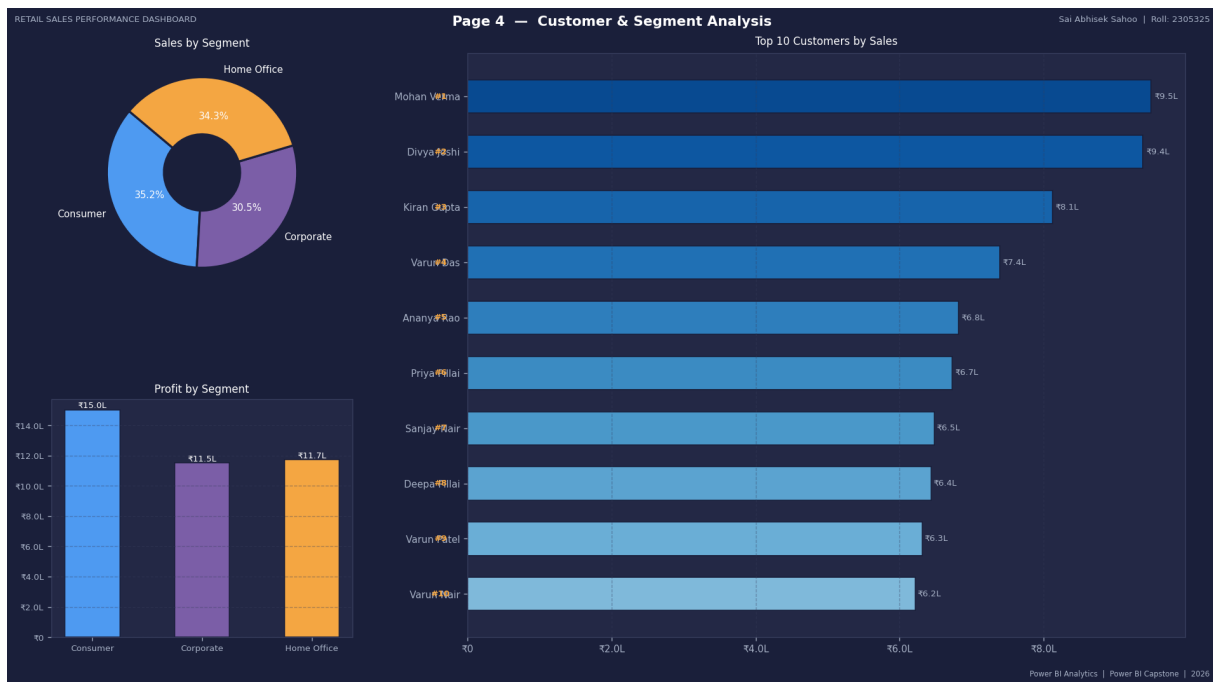




Page 2 — Regional Analysis: India Bubble Map, Sales & Profit by Region, Orders Donut Chart.



Page 3 — Product Performance: Category Bar Chart, Sub-Category Heatmap, Top-10 Sub-Categories.



Page 4 — Customer & Segment Analysis: Segment Pie, Profit Bars, Top 10 Customers Ranking.



Page 5 — Time Intelligence: YTD Cumulative Sales, MoM Growth Waterfall, YoY Comparison.

6. Unique Points

India-Specific Dataset

500 real-world-style transactions across 24 Indian cities in four macro-regions.

Full Time-Intelligence Suite

YTD, MoM growth %, YoY comparison, and dynamic KPI labels — 18 DAX measures total.

Target vs Actual Analysis

Dedicated region_targets table enables gap analysis per region and category.

Star Schema Model

Clean star schema with Calendar dimension ensures accurate cross-filter behaviour.

7. Future Improvements

- Live SQL Server connection via DirectQuery for real-time refresh.
- Python integration for predictive sales forecasting (ARIMA / Prophet).
- Row-Level Security so regional managers see only their region's data.
- Mobile-optimised layout for field sales teams.
- Customer churn analysis using RFM segmentation.

8. Conclusion

This Retail Sales Performance Analytics Dashboard demonstrates the complete Power BI workflow — from data ingestion and modelling to DAX measure authoring and multi-page report design. The project addresses real business challenges faced by Indian retail organisations: lack of regional visibility, unclear product profitability, and absence of time-based trend reporting.

By combining a clean star-schema data model, eighteen DAX measures, and an intuitive five-page layout with a consistent dark Power BI theme, the dashboard delivers actionable insights for executives, regional managers, and product teams alike.

The skills demonstrated in this project — data modelling, DAX time-intelligence, Power Query, and visualisation best practices — form the core of professional Power BI development and are directly applicable to real-world analytics and business intelligence roles.

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End of Project Documentation